

Week 6 Mandatory Hand-in: Open Economy, Exchange Rates and the Peg

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Purpose. This assignment covers the main mechanisms from Lectures 9 and 10: uncovered interest parity, the trilemma, real exchange rates, the Marshall–Lerner condition, and adjustment under a fixed exchange rate. The central question is: how does a small open economy adjust when the nominal exchange rate is not the adjustment margin?

Exam qualification. This is a mandatory pass/fail hand-in. You may work in groups of at most 4 students. Use the notebook for the programming parts and write short Markdown answers inside the notebook. Submit your completed notebook, renamed `week06_open_economy_GROUPNAME.ipynb`.

Files.

- Student starter notebook: `week06_open_economy_student_start.ipynb`
- No separate data file is needed; the notebook contains a small stylised Denmark/euro-area inflation panel.

What counts as a pass? A passing submission must:

1. run from a clean kernel without errors after you have completed the TODOs;
2. contain the requested tables and figures with readable labels;
3. include short interpretations after the main calculations;
4. correctly use UIP, the real-exchange-rate identity, Marshall–Lerner logic, and the fixed-rate AS–AD model;
5. end with a concise policy memo that refers to your own numerical results.

Notation. The nominal exchange rate is the domestic-currency price of one unit of foreign currency. Thus E rising means domestic depreciation. In the analytical questions, interest rates and inflation rates are written in percent unless explicitly stated. In the Python notebook, use decimals for rates unless the cell tells you otherwise.

Part 1. UIP, credibility and the trilemma

Let the foreign nominal interest rate be $i^f = 2.5\%$. The exact UIP condition is

$$1 + i = (1 + i^f) \frac{E_{t+1}^e}{E_t},$$

and the log-linear approximation is

$$i \approx i^f + \Delta e_{t+1}^e.$$

Treat expected percentage changes as small log changes.

1. Derive the log-linear UIP equation from the exact condition.
2. Compute the domestic nominal interest rate using both the exact and the log-linear formula in the following cases:

Scenario	Probability of devaluation	Devaluation size
Credible peg	0	0
Mild speculative pressure	20%	6%
Severe speculative pressure	40%	10%
Expected appreciation	–	–1%

For the devaluation cases use $\Delta e^e = p\delta$.

3. A large neighbouring central bank cuts its policy rate by 0.5 percentage points. The small economy keeps a credible peg and has free capital mobility. What should happen to the small economy's policy rate? Explain using the trilemma.
4. Suppose markets assign probability $p = 0.30$ to a devaluation of size $\delta = 8\%$. If $\beta_2 = 0.40$ measures the effect of a one-percentage-point increase in the real interest rate on output demand, compute the immediate demand shift from the expected-devaluation premium. Explain the economic mechanism in one paragraph.

Part 2. Real exchange rates, competitiveness and Marshall–Lerner

The real exchange rate is

$$E_t^r = \frac{E_t P_t^f}{P_t}, \quad e_t^r = e_t + p_t^f - p_t.$$

A higher E^r means that domestic goods become cheaper relative to foreign goods, so domestic price competitiveness improves.

1. Show that

$$\Delta e_t^r = \Delta e_t + \pi_t^f - \pi_t.$$

2. Under a fixed nominal exchange rate, $\Delta e_t = 0$. Compute Δe^r in the following two cases and state whether competitiveness improves or worsens:

(a) $\pi = 1.2\%$, $\pi^f = 3.0\%$;

(b) $\pi = 3.5\%$, $\pi^f = 2.0\%$.

3. Starting from balanced trade, a real depreciation changes net exports approximately by

$$\frac{\Delta NX}{E^r M} \approx (\varepsilon_X + \varepsilon_M - 1) \frac{\Delta E^r}{E^r}.$$

For a 5% real depreciation, compute the effect for the following elasticity pairs and say whether Marshall–Lerner holds:

$$(\varepsilon_X, \varepsilon_M) \in \{(0.4, 0.3), (0.7, 0.5), (1.2, 0.6)\}.$$

4. Suppose the long-run elasticity sum is $\varepsilon_X + \varepsilon_M = 1.2$, but in the first quarter only 20% of the quantity response has occurred. Compute the first-quarter effect and the long-run effect of a 5% real depreciation. Explain the J-curve in two or three sentences.

Part 3. The fixed-rate AS–AD model

Work with the fixed-rate model in deviations from long-run equilibrium:

$$\hat{y}_t = \beta_1(q_{t-1} - \hat{\pi}_t) + z_t, \quad (\text{AD})$$

$$\hat{\pi}_t = \gamma \hat{y}_t + s_t, \quad (\text{AS})$$

$$q_t = q_{t-1} - \hat{\pi}_t. \quad (\text{RER})$$

Here q_t is the real-exchange-rate gap and $\hat{\pi}_t = \pi_t - \pi^f$. Use $\beta_1 = 0.7$ and $\gamma = 0.5$ for numerical answers.

1. Solve the AD and AS equations for $\hat{y}_t$ and $\hat{\pi}_t$ as functions of q_{t-1} , z_t and s_t .
2. Set $z_t = s_t = 0$ and show that

$$q_t = \frac{1}{1 + \beta_1 \gamma} q_{t-1}.$$

Explain why this implies stability. If $q_0 = -2$, compute $\hat{y}_1$, $\hat{\pi}_1$ and q_1 .

3. Consider a temporary fiscal expansion: $z_1 = 1$, $z_t = 0$ for $t \geq 2$, $s_t = 0$, and $q_0 = 0$. Compute $\hat{y}_1$, $\hat{\pi}_1$, q_1 and $\hat{y}_2$. Explain why output can fall below potential after the stimulus has ended.
4. Consider a temporary adverse supply shock: $s_1 = 0.8$, $s_t = 0$ for $t \geq 2$, and $q_0 = 0$. Compute $\hat{y}_1$, $\hat{\pi}_1$ and q_1 . Explain the signs.

Part 4. Python simulation

Complete the starter notebook. Your notebook should contain the following outputs:

1. A UIP table comparing the exact and log-linear domestic interest rate for the four cases in Part 1, plus a plot of both UIP schedules for expected depreciation from -2% to 6% .
2. A Marshall–Lerner table and the first-quarter versus long-run J-curve calculation from Part 2.

3. A plot of the stylised Denmark/euro-area inflation differential and the implied real-exchange-rate index under a fixed nominal exchange rate. Use the notebook's approximation

$$q_t = q_{t-1} + \frac{\pi_t^f - \pi_t}{12}.$$

Note. The q_t in this monthly approximation is a real-exchange-rate *level index* (Jan 2021 = 100), built from the identity $\Delta e_t^r = \pi_t^f - \pi_t$ rescaled to a monthly increment. It is not the same object as the model gap q_t used in Part 3 and Part 4(4), which is a deviation from long-run equilibrium.

4. A simulation of the temporary fiscal expansion in the fixed-rate AS–AD model for $T = 20$. Plot $\hat{y}_t$, $\hat{\pi}_t$ and q_t .
5. A comparison of $\beta_1 \in \{0.4, 0.7, 1.0\}$. For each value, report the convergence root $1/(1+\beta_1\gamma)$ and the half-life $\log(0.5)/\log(\text{root})$.
6. A comparison between the fixed-peg model and the closed-economy AS–AD model with static expectations,

$$\hat{y}_t = z_t - a\hat{\pi}_t, \quad \hat{\pi}_t = \hat{\pi}_{t-1} + \gamma\hat{y}_t,$$

using $a = 1.2$, $\gamma = 0.5$ and the same temporary demand shock $z_1 = 1$. In this closed-economy block $\hat{\pi}_t$ denotes the deviation of inflation from the steady-state target (there is no foreign anchor here); static expectations means $\pi_t^e = \pi_{t-1}$, which generates the lagged term in AS. The slope a is the reduced-form sensitivity of demand to the inflation gap (it bundles the real-rate channel and a Taylor-type monetary response).

After each figure, add a short interpretation. Do not write long paragraphs; two or three precise sentences are enough.

Part 5. Policy memo

Write 250–350 words. Suppose Denmark is hit by a domestic recession while the euro area is close to normal. The peg is credible at first, but markets start to discuss a possible devaluation. Should policy rely on fiscal expansion, wait for relative-price adjustment, or consider devaluation? Your answer must mention:

1. UIP and the trilemma;
2. the real-exchange-rate adjustment channel under a peg;
3. the temporary-fiscal-stimulus “competitiveness hangover” from your simulation;
4. why an anticipated devaluation can be contractionary before it happens.